EFFECT OF LOAD AND REPETITION RANGE ON BICEPS MUSCLE ACTIVATION DURING BICEP STIMULATION

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INTRODUCTION

Currently in the gym and athletic community there are lots of debates over which curl variations, loads, and ranges of motion (ROM) output the biggest bicep activation and hypertrophy. This topic has caused disagreements because changes in elbow angle, shoulder position, repetition tempo or weight all impact on how the elbow flexors generate their force. According to a study on the differences in activation patterns in biceps brachii (BB) during isometric, concentric and eccentric contractions it states that, "The multi-channel sEMG variables differed between the contraction types (isometric, concentric and eccentric) during MVCs reflecting differences in respective motor control strategies." [1] This means the slightest change results in a shift in the muscle recruitment patterns. To be more specific, these disagreements are mainly due to the redundancy of the human motor system since, "dynamic EMG data show that smooth extension of the elbow joint requires a fine interplay between the flexors and extensors." [2] As a result, previous studies are showing that "the relative activation between the BB and the brachioradialis (BR) muscles depended on the angle of the elbow joint." [3] Due to these arguments, it is important to conduct a research study to compare how different loading strategies such as low repetition, high weight curls versus a controlled moderate weight condition affect BB activation.

The main objective of this study was to measure and compare normalized EMG activation in the long and short heads of the BB and BR through a standardized control condition and a low repetition, high weight condition. In addition, this study was trying to determine if increased weight resulted in measurable form deterioration by analyzing shoulder and elbow joint kinematics. It was hypothesized that lengthened joint angles would produce the highest activation, heavier weight would increase EMG peaks relative to the control, and that greater task difficulty would lead to form deterioration and a shift in the motor strategy. The last hypothesis states that higher and more consistent EMG activation will result in more hypertrophy. Overall the purpose of this study was to determine if the low repetition, high weight bicep curl would produce higher activation than the controlled condition at different elbow angles and form deterioration with heavier weights.

METHODS

One healthy male participant (P1: age 20, height: 183cm, and 74kg) reviewed and provided written informed consent approved by the University of Guelph Research Ethics Board before any testing was done. Participant wore a short sleeve shirt which allowed EMG to be placed on the upper arm without any obstruction to shoulder and elbow markers.

Reflective markers were placed on the upper limb by using the HASMotion model 1 (HASMotion Inc., Kingston, ON, Canada) marker configuration. This was done to help compute elbow and shoulder joint angles on Visual3D (Professional 2025, HAS-Motion, Kingston, ON, Canada) to help discover form deterioration throughout the experiment. The EMG electrodes were placed on the BB short head (channel 1), BB long head (channel 2), and BR (channel 3) of the participant's dominant arm. These markers were placed according to the SENIAM guidelines. Before marker placement, each site was shaved, lightly abraded, and cleaned with alcohol to minimize any discrepancies in data. All electrodes and placement of the markers were photographed with consent from the participant for documentation purposes. All kinematic data was collected using the VICON Motion Capture System (v2.16, Vicon Motion Systems Ltd., Oxford, UK) and the wireless Delsys Trigno surface EMG system (Trigno, Delsys, Natik, MA, USA). With VICON Nexus being open on a computer that is synced with the EMG system and VICON motion capture to sampling rates set at 2000 Hz for consistent high frequency EMG content. Once VICON motion capture

system was fully calibrated using the L-frame and wand, a short test was run to ensure marker visibility, correct labelling, and if the EMG readings did not have excessive noise.

Before any main trials were conducted, participant completed a warmup using lightweight dumbbells that require very little muscle activation to familiarize themself with the movement for actual data collection. Participant then conducted a one repetition maximum (1RM) of 35 pounds for the dumbbell bicep curl using the incline bench and weight adjustable dumbbell and a standard incremental loading approach. After this 1RM, the participant conducted two maximal voluntary contraction (MVC) trials to use as a 100% muscle activation benchmark for when the EMGs were normalized. Throughout the MVCs, the participant produced maximal isometric elbow flexion against manual resistance from a fellow lab partner for about 3 seconds with 2 to 3 repetitions all separated with rest.

All dynamic trials were performed on an adjustable incline bench set to approximately 45°. There were other conditions performed throughout this experiment although the condition analyzed in this report was done first so there will be very minimal fatigue. For each trial, participant was kneeling on the floor with their dominant arm laying supinated with a dumbbell (supinated grip) against the inclined bench (Similar to preacher curl). Between each set, a rest period was given to minimize fatigue and keep consistent data throughout the trials. A metronome was used to cue 2 second concentric and eccentric tempo.

For the control condition, participant performed 8-10 repetitions with a dumbbell around 50% (17.5 pounds) of the 1RM. For the first condition (low repetitions, high weight), the participant performed 6 to 8 reps using around 75% (25 pounds) of the 1RM. For both conditions, participant started each repetition with a fully extended elbow while maintaining contact with the bench. Then participant curled through their comfortable range of motion and then returned back to the starting position to complete one rep. Any trials with obvious tempo errors, loss of contact with the bench, or incorrect rep count were redone to maintain consistency in data.

All VICON motion capture files were saved and exported using a C3D format and respective EMG data was exported as .csv files. The kinematic and EMG data was processed using VICON Nexus, Visual 3D, and MATLAB. The shoulder and elbow joints were analyzed and gap filled within Visual 3D and VICON Nexus and exported for further processing on MATLAB to identify form deterioration. Lastly, within MATLAB, the EMG signals were processed using a standard pipeline consisting of band pass filtering, full wave rectification, and low pass filtering to obtain a clear and smooth presentation of the muscle activation over time. Each muscle channel was normalized to their respective peak MVC value to show the activation in terms of %MVC.

RESULTS

After analyzing and processing all the data, the muscle activation was very different between the control and condition (low repetition, high weight) trials. The condition was consistently producing much higher values. To be more specific, the condition shows a greater peak %MVC for all muscles, with an increase of approximately 89% for the BB short head (220.5 %MVC compared to 130.6%MVC), 31% increase for the BB long head (227.8%MVC compared to 173.3%MVC), and 44% increase in the BR (144.1%MVC compared to 100.1%MVC) (Figure 1 and Table 1).

The EMG time related data showed a similar trend in terms of the condition by having higher amplitudes in the bursts for every muscle. This pattern also shows that there was consistent elevated activation instead of just a single isolated peak (Figure 2). When the activation analyzed relative to the elbow joint angle, both the control and condition trials showed an early activation peak in EMG amplitudes when the elbow was in a lengthened position of approximately 20° to 40°. Although, the condition showed a much higher activation around 30° for all heads of the BB and the BR (45-47%MVC long head and approximately 38%MVC for BR and short head) compared to the lower peaks of the control trial (Figure 3). After the mid range angles, the activation began to generally decline for both trials, but the condition still had a higher amplitude throughout the whole shift.

In terms of shoulder kinematics, they remained stable throughout the control condition with an average of approximately -43° across the repetitions. In contrast, the condition trial showed a consistent increase in the shoulder flexion going from –8.8° to –12.2°. Considering this there was approximately a 30° to 35° shift in shoulder position relative to the control (Table 1). Throughout the condition there resulted a 3.4 total drift in the shoulder angle (x direction) from the start and end of the experiment while the control only showed a 1° total drift (Table 1).

Rep	Control (°)	Condition (°)	Difference (°)
1	-43.1	-8.8	34.3
2	-43.9	-10.3	33.6
3	-43.8	-9.2	34.6
4	-43.2	-11.9	31.3
5	-43.7	-11.5	32.2
6	-43.3	-11.5	31.8
7	-43.6	-12.2	31.4
8	-43.3	-12.3	31
9	-44.1	-12.3	31.8
10	-43.6	-13.3	30.3

Table 1: Shoulder joint angles (X-direction) across ten repetitions for control versus the condition trial, showing increased shoulder extension during high-load repetitions.

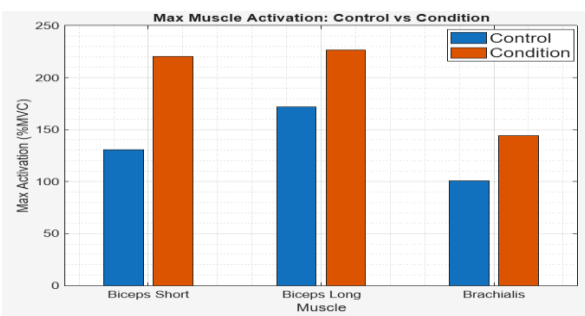

Figure 1: Peak normalized EMG (%MVC) for each elbow-flexor muscle between the control and condition trials, highlighting the increased activation with higher weight.

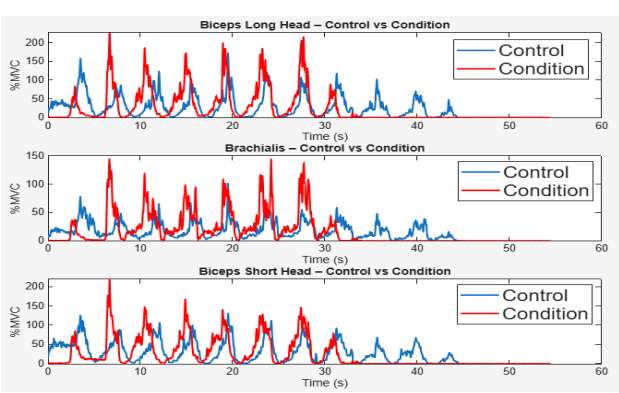

Figure 2: Normalized EMG (%MVC) for the biceps long head, brachialis, and biceps short head across all repetitions, comparing the activation bursts between the control and condition trials.

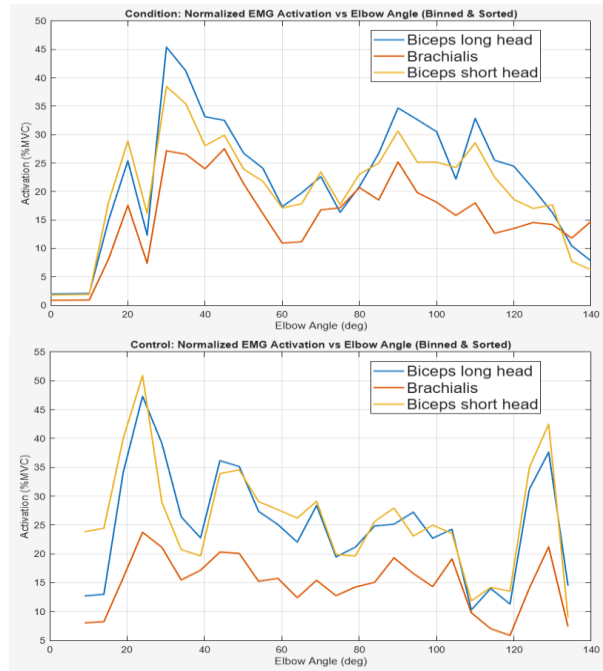

Figure 3: Normalized EMG activation of the BB long head, BB short head, and BR across elbow angle for both the control and condition trials, binned and sorted to show angle-dependent activation patterns and differences in peak activation between loading conditions.

DISCUSSION

The condition trial consistently showed a higher muscle activation than the control. This means there was a higher neuromuscular demand under the increased loading. This is proved in the fatigue literature which explains how repetitive contractions cause "changes at biochemical, systemic, and structural levels that compromise activity" [4]. This explains the downfall in muscle activation later in the condition trial. The activation in the early repetitions also shows the recruitment of more motor units since the fatigue studies show that, fatigue contractions "increase recruitment of additional high threshold motor units to preserve the desired force level" [5].

Although the angle dependent muscle activation graphs the amplitudes peaked around 20° to 40°, this is highly likely due to inertial loading. The activation increases again near the mid range angles where a study proved that, "manipulating joint angle and

training at different muscle lengths...impact the acute responses associated with EMG" [6]. This is consistent within resistance training research where, "the manipulation of resistance training variables is paramount for maximizing muscular adaptations" [7]. This proves that altering the joint angle directly has an effect on the mechanical tension and recruitment. In addition, more evidence states that resistance training performed near failure can result in, "a similar degree of muscle fiber activation to that of high intensity regimes" [8]. This is proved through the high activation observed within the condition even though there were less repetitions. Therefore, the higher activation shown in the condition proves that the muscle had to recruit more motor units because of the heavier load and the increased effort in the eccentric movement. Considering this, the higher and more consistent the muscle activation in the condition hence proves the hypothesis that higher consistent EMG data is directly related to more hypertrophy. During the condition trial, the shoulder angle was displaced by around 3.4° since using heavier weights shows that, "the dynamic system experiences continuous status transitions that culminate in task failure" [9]. This is why there was an increase in the shoulder angle on the fourth repetition and after. Thus, the shoulders were used and resulted in a shift in the motor strategy in the condition trial. In other words, the condition trial showed a consistent increase in extension which supports the resistance training research that, "high load training is superior for enhancing muscle strength" [10]. This proves that with higher weights and more neuromuscular demand, there will be an additional need for more muscle groups to be used which results in form deterioration. Considering all of this, the data supports the hypothesis that higher task difficulty results in measurable form deterioration.

Key limitations can include that the shoulder angles were not fully accurate since in Visual3D the reference segment was used as the empty lab space rather than the anatomical segments. Another limitation is results for hypertrophy due to the short experimental timeframe given. Future work could improve accuracy by incorporating a thorax/ab model using HASMotion guidelines, using more EMG placements on the shoulder to get more accurate results, and assessing hypertrophy over time.

CONCLUSION

The results supported the hypotheses by showing consistent higher activation in the condition, evident angle dependent activation patterns, measurable form deterioration with increased weight, and higher hypertrophy due to more consistent EMG activation. In the future, it would be beneficial to improve shoulder angle modelling within Visual3D, use more EMG sensors for more accurate deterioration results, and assess hypertrophy over a longer time period.

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